

SimpleNotebook

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0.1 Jupyter Notebook with Integrals

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0.3 SciML - Fall 26

$$\int_{-1}^1 \frac{1}{1+x^2} dx$$

```
[2]: import scipy.integrate as integrate
      # Define the function
      def f(x):
          return 1 / (1 + x**2)

      # Compute the definite integral from -1 to 1
      result, error = integrate.quad(f, -1, 1)

      # Print the result
      print(f"Integral result: {result}")
      print(f"Estimated error: {error}")
```

Integral result: 1.5707963267948968

Estimated error: 1.7439342485646153e-14